📘 C3 Architecture Overview v2.0.0-alpha

C3 Contract Control Center

Architecture Overview

Version: 2.0.0-alpha

Status: Current Architecture

Last Updated: 26 June 2026

Purpose

This document provides a high-level overview of the C3 Contract Control Center architecture.

It serves as the primary reference for understanding how application components interact, how business logic is organized, and how data flows throughout the platform.

Unlike Architecture Decision Records (ADRs), this document describes the current implementation rather than documenting individual architectural decisions.

Architectural Philosophy

C3 follows a layered architecture designed around separation of responsibilities.

Each layer has a clearly defined purpose and communicates only with the layer immediately below it.

The objective is to maximize maintainability, testability, and long-term scalability while minimizing coupling between infrastructure and business functionality.

High-Level Architecture

Presentation Layer

Contracts

People

Renewals

Amendments

Inbox

Intelligence

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Application Layer

React Hooks

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Service Layer

SPService Interface

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Adapter Layer

Mock Adapter

SharePoint Adapter

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Translation Layer

Contract Mapper

Person Mapper

Amendment Mapper

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Domain Layer

DTOs

Business Rules

KPI Engine

Operational Insights

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Data Source

SharePoint

Mock Data

Future APIs

Platform Layers

Presentation Layer

Responsible for rendering user interfaces.

Contains:

- Workspaces
- Profile pages
- Dashboards
- Shared UI components

Responsibilities:

- Display information
- Handle user interaction
- Invoke React Hooks

The Presentation Layer contains no business rules.

Application Layer

Implemented through React Hooks.

Examples:

- useContracts()
- useContract()
- usePeople()
- usePerson()
- useAmendments()
- useIntelligence()

Responsibilities:

- Data orchestration
- Loading state management
- Error handling
- Composition of domain services

Service Layer

Defines the stable application contract.

Responsibilities:

- Abstract data retrieval
- Hide infrastructure details
- Expose domain-oriented operations

The Service Layer represents the only approved entry point into platform data.

Adapter Layer

Provides infrastructure implementations.

Current adapters:

- Mock Adapter
- SharePoint Adapter

Future adapters may include:

- REST APIs
- Dataverse
- Platform Services

Adapters satisfy the Service Layer contract.

Translation Layer

Transforms infrastructure objects into domain models.

Current mappers:

- Contract Mapper
- Person Mapper
- Amendment Mapper

Responsibilities:

- Normalize SharePoint payloads
- Resolve lookup fields
- Handle null values
- Produce deterministic DTOs

Domain Layer

Contains application business knowledge.

Includes:

- Domain DTOs
- KPI calculations

- Renewal logic
- Operational Insights
- Intelligence aggregation

This layer remains completely independent of SharePoint.

Intelligence Architecture

Operational intelligence is treated as a first-class platform capability.

Current intelligence components include:

- Contract KPIs
- Renewal KPIs
- Workflow analysis
- Amendment activity
- Team distribution
- Game distribution
- Operational Insights

Dashboard components consume precomputed intelligence rather than implementing business calculations directly.

Architectural Principles

The current architecture is governed by the following principles:

- Presentation contains no business rules.
- Hooks orchestrate rather than calculate.
- Services define stable contracts.
- Adapters isolate infrastructure.
- Mappers isolate storage models.
- DTOs represent business entities.
- Intelligence centralizes analytical logic.

Current Domains

The platform currently implements:

- Contracts
- People
- Amendments
- Renewals
- Inbox
- Intelligence

Future domains may expand this architecture without requiring structural redesign.

Current Status

Architecture Status: Stable

Phase 2 established the complete architectural foundation for the C3 Contract Control Center.

Future development focuses on:

- Live SharePoint integration
- Power Automate write operations
- Authentication
- Production deployment
- Platform expansion

Summary

C3 is designed as a layered operational platform rather than a collection of independent pages.

Its architecture intentionally separates presentation, orchestration, business logic, infrastructure, and storage to ensure long-term maintainability and scalability.

The platform is considered architecturally stable as of **v2.0.0-alpha**.
